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Fecha Lun 20/04/2026 21:43

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Dear JAIME A CUCHIA,

On behalf of the MISNC 2026 Program Committee, we are pleased to inform you that your paper has been accepted for presentation at MISNC 2026. The conference will be held on August 10-12, 2026, in Da Nang, Vietnam.

Paper Information

Paper ID: 81

Title: A Comparative Analysis of Machine Learning Models Employed for Spatiotemporal Trajectory Prediction in BRT Systems

Track: Special Session on Network, Economic and Business Studies

Authors: JAIME A CUCHIA

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If you have any questions, please do not hesitate to contact us.

We look forward to welcoming you this summer in Da Nang, Vietnam.

Best regards,

MISNC 2026 Program Committee

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A Comparative Analysis of Machine Learning Models Employed for Spatiotemporal Trajectory Prediction in BRT Systems

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Abstract: This study develops the first predictive framework for passenger demand at 15-minute intervals for the TransMilenio BRT line (Troncal 80) in Bogotá, using the largest dataset to date—with nearly 500,000 records—that combines spatial inputs, timestamps, operational data, and historical trends. The authors also took extreme care to segment their data into training, validation, and test sets so that they could avoid leakage and bias, especially in their comparisons of various machine learning models (Random Forest, XGBoost, SVR, MLP, and LSTM).

For example, the MLP neural network model produced the most favorable results in terms of the lowest error rates (MAE 0.011, RMSE 0.043, $R^2 \sim 1$, MAPE < 0.1%), and also in its extremely low inference time of 4.8 microseconds per prediction, which is particularly relevant for real-time adjustments in smart city operations. XGBoost was nearly indistinguishable from the MLP in overall performance (R^2 0.9986), and even provided some transparency regarding the contribution of features. Among the predictors, the inflow from previous time intervals in the time series model was more significant than the multiple static models. Consequently, this study can now provide other cities in Latin America (including Bogotá) and other metropolitan regions around the world with an initial framework for demand forecasting.

Keywords: Machine Learning; Bus Rapid Transit (BRT), Spatiotemporal trajectory forecasting; Random Forest, XGBoost; Support Vector Regression (SVR), Multilayer Perceptron (MLP), Long Short-Term Memory (LSTM), Temporal validation.

1 Introduction

In Latin America, the most common way to get around is by Bus Rapid Transit (BRT) systems. They provide metro-level service capacity at a significantly reduced cost for the infrastructure [1]. In this case, Bogotá D.C. The TransMilenio (TM) system has been around since the early 2000s and is now one of the most well-known BRT systems in the world. TM now has 143 stations, 9 portal terminals, and more than 114.4 kilometers of trunk lines that are only for them. There are about 4,000 articulated, bi-articulated, and feeder vehicles in the fleet that move about 4.03 million people every day. Because TransMilenio is so big, it's hard to plan services for it. You need to know how many people will be at each station every 15 minutes. This is crucial for optimizing operations, ensuring real-time fleet mobility, and minimizing wait times for passengers [2]. Models that predict challenges, despite the fact that BRT systems in Latin America face issues

related to the temporal granularity of the systems, the integration of system-specific operational components, and rigorous validation [3], [4].

The gap between what needs to be done and what is technologically feasible drives the creation of more reliable, situation-specific forecasting systems. To overcome this, it is necessary to analyze the problems from two different angles. The temporal perspective refers to time-dependent demand variability, manifested through cycles—whether by time of day, day of the week, or even specific weather conditions. On the other hand, the spatial perspective captures the net movement of passengers between stations along the corridor—which a priori aggregate models overlook—and exhibits structural variability.

The prediction of spatiotemporal trajectories arises from the synthesis of both viewpoints [2]. In this case, Machine Learning models are useful because they can find non-linear relationships between covariates without making strict assumptions about how the data was generated [3]. Recent research in BRT and metro systems in Asia and Europe have shown that gradient boosting and neural network algorithms work better than traditional statistical methods [4], [5]. Still, there isn't enough proof about conditions in Latin America. because there isn't enough operational data that is broken up into pieces [6].

This work presents four new ideas because of this situation. It makes a dataset that has never been seen before of TransMilenio Troncal 80 has 493,662 operating recordings, and each one can tell you the time to the nearest 15 minutes. The authors say that this dataset is the best one for a BRT corridor in Latin America. It also does a systematic comparative analysis of five Machine Learning models across three principal algorithmic categories, conducted within a stringent temporal validation framework that safeguards against future data leakage. Third, it separates predictive performance into 13 corridor stations and four time periods for a study of space and time. Finally, it tells you how much it costs to run each model at the moment. This work has four parts. (1) *Data and methodology* (2) results, (3) starts a conversation, and (4) puts the findings together and suggests new ways to do research.

2 Data and methodology

2.1 Data Acquisition, Cleaning, and Integration

The data was collected from the TransMilenio open data portal [10], and bus tickets were validated using two data sources: passenger entry data by station and time range; and passenger exit log data. These sources served as the basis for 28 monthly spreadsheets containing approximately 58,297,600 records. Because automated fare collection (AFC) systems are known in the literature [11, 12] to be plagued by multiple issues—such as missing entries due to turnstile failures, duplicates from transmission errors, duplicates from operational disruptions, or outliers caused by operational disruptions—we created a 6-step process to clean the data: (i) we filtered by the 80th Street corridor (TZ005) to leave 2,961,104 records (representing 5.1% of the total) [10]; (ii) we restricted the data temporally to the period from January 2025 to February 2026 [13]; (iii) we removed duplicates using the composite key (station, date, interval) [16]; (iv) we performed data imputation to fill in missing values in key variables [3],[18]; (v) we identified outliers in the data using the IQR method for each station, taking into account the natural variability in demand for the 13 stations [4, 14]; and (vi) we ensured that the temporal coverage of the data for each station was complete during the reviewed period [18]. The process retained 91.6% of validation records

(1,302,310 $\rightarrow$ 1,192,697) and 92.3% of exit records (1,658,794 $\rightarrow$ 1,530,724), with no loss of station or time coverage. The resulting demand distribution and hourly profile are shown in Figs. 1 and 2. The cleaned data were then integrated with structural GTFS sources through a spatiotemporal join on the composite key, incorporating scheduled headway as a supply proxy in the absence of real-time fleet telemetry [2]. Integration with the TransMilenio 2026 GTFS feed produced median headway values ranging from 96 s (Carrera 77, a high-frequency node) to 212 s (Carrera 53 and Escuela Militar), yielding a final integrated dataset of 493,662 unique records across 20 columns [15].

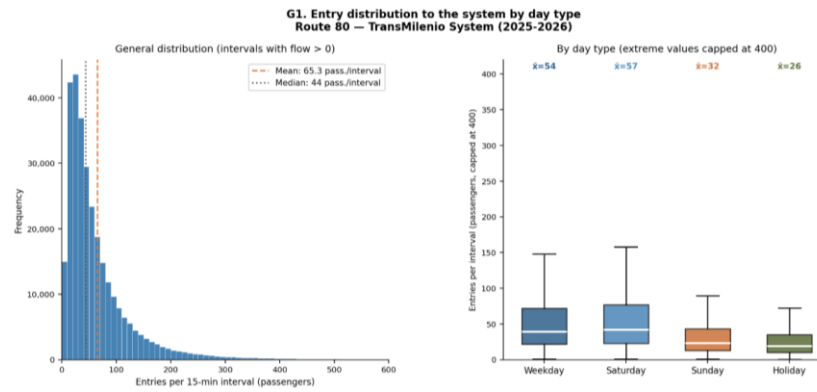


Fig.1. Distribution System By: Day type – Route 80 2025-2026 (Own Elaboration)

Figure 2 shows the average hourly inflow pattern, broken down by day of the week on the left and by station on weekdays on the right. The profile follows the typical morning and evening peak structure commonly seen in urban BRT systems, as observed along Troncal 80 during 2025–2026.

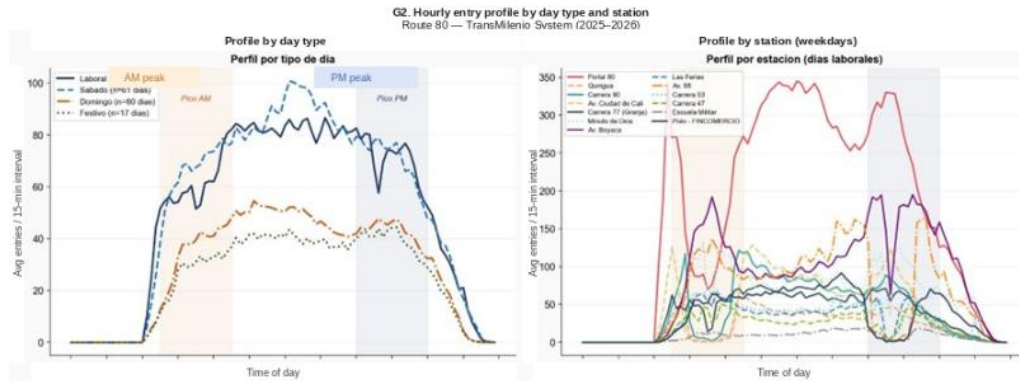


Fig.2. Hourly entry profile by: date type and station (Own Elaboration)

2.2 Variable Engineering and Preprocessing

Twelve input variables were selected using a preliminary importance analysis with Random Forest (See Table 1) [7]. Categorical variables were encoded using *Label Encoding* supervised by frequency of occurrence; all variables were standardized using *StandardScaler* trained exclusively on the training set, ensuring that the validation and test sets are only transformed—never trained—to prevent information leakage [8]. The goal is to find out how many Every 15 minutes, the number of people getting on and off at each station.

We used Random Forest to find the twelve most important features (see Table I).

Table 1. Input Features

Variable	Type	Description
Encoded station ID	Spatial	Frequency-encoded station code (0–12)
Ordinal corridor position	Spatial	Station position along the corridor (0–12)
15-min interval index	Temporal	Interval number within the day (0–95)
Day of week	Temporal	0 = Monday through 6 = Sunday
Day-type encoding	Temporal	Weekday, Saturday, Sunday, public holiday
Time-period encoding	Temporal	Night, AM peak, off-peak, PM peak
Passenger egress	Operational	Alighting passengers per interval
Net passenger flow	Operational	Boardings minus alightings
Boarding/alighting ratio	Operational	Capture vs. dispersal at each station
Previous-interval boardings	Autoregressive	Boardings in the immediately preceding interval
Scheduled headway	Operational	Planned gap between consecutive buses (seconds)
Vehicle capacity	Operational	Passenger capacity of the assigned vehicle

2.3. A time-based strategy for validating

A rigorous approach for time validation was employed to replicate actual deployment scenarios [9], [8]: training set. The validation used 315,891 records from January to September 2025 and 70,608 records from October to November 2025. choosing hyperparameters; and a test set with 107,163 records from December 2025 to March 2026, just for last test. This strategy guarantees that learning isn't affected by subsequent observations. Cross-validation, on the other hand, mixes together observations at random and allows time-sensitive data leak [10]. . We used RandomizedSearchCV with three time folds on the training set to discover the optimum hyperparameters [11]. MAE, RMSE, R^2 , and MAPE are all approaches to measure how well something works.

2.4 Machine Learning Models: Fundamentals and Configuration.

The three families of machine learning algorithms selected to predict transportation demand are artificial neural networks, tree-based ensemble methods, and kernel regression.

For the Random Forests (RF) model, 200 decision trees were constructed. Each tree was trained using a bootstrap sample, and a random subset of features ($\log_2$ of the total) was used at each split. Predictions are made by averaging the predictions from all trees, rather than relying on a single one, which reduces variance without increasing bias. Since predictions are made using the average of all trees, RF is robust to noise and inconsistencies in the AFC validation data. The RF model underwent variable importance evaluation using Mean Decrease in Impurity (MDI). Each feature was assigned a normalized score and a ranking based on how much it contributed to

the demand pattern. The trees grew without depth constraints, allowing the ensemble mechanism to fully control overfitting [15], [16].

RF and XGBoost (XGB) differ in that RF builds multiple trees in parallel, while XGB builds trees one at a time sequentially. Each successive tree is built to correct the errors of the previous tree. This tree-building method makes XGB suitable for the Troncal 80 corridor, where demand can vary by a considerable order of magnitude between adjacent stations. Regularization (learning rate $\lambda = 0.1$) was used to ensure that the model did not overfit too quickly. Importance was measured using normalized gain. Following [7], the model was built with 500 estimators, a maximum depth of 6, and $\gamma=0.2$.

With SVR and its RBF kernel, rather than partitioning the feature space, a predictive function is fitted within a tolerance range (ϵ) around the training targets, and the support vectors are the only points outside this range that determine the prediction. The RBF kernel captures nonlinear relationships in demand, as the kernel function measures similarity using Euclidean distance in a standardized 12-dimensional space. A model with $C=500$ means that the model's deviation is minimal, and given the characteristics of the dataset, this is appropriate. Considering the computational cost of SVR, training was performed on a smaller subset of the data, 80,000 records; in transportation demand forecasting, sensitivity analysis suggests that 20–25% subsampling adequately reflects the distribution of the original dataset when standardization is applied [17]. The chosen configuration was $C=500$, $\gamma=0.1$, and $\epsilon=1.0$.

The MLP predicts passenger flow using 12 input neurons, a single hidden layer with 128 neurons using ReLU, and 1 output neuron. The MLP was trained using backpropagation (BP) with the Adam optimizer to minimize the MSE (related to passenger flow) with L2 regularization ($\alpha=0.01$) and early stopping as strategies to avoid overfitting. Despite its simple structure, the model was able to provide useful predictions as a result of domain knowledge.

Finally, LSTM was selected because it learns sequential dependencies using internal cell states and has three trainable gates—forget, input, and output—that control the flow of information and address the vanishing gradient problem in standard RNNs. The model uses a sliding window of 8 consecutive 15-minute intervals (2 hours of data) to predict the next arrival. The architecture has 64 hidden units, a dropout rate of 0.2, 32 fully connected units with ReLU, and a single-unit output layer. Training used Adam with a learning rate of 1×10^{-3} and early stopping. All of this was implemented in PyTorch 2.5.1 [34] and ran on an NVIDIA RTX 3060 with 12 GB of VRAM. Table 2 summarizes the model specifications.

Table 2. Model Configurations and Hyperparameters (Own Elaboration).

Model	Hyperparameters	Search	Hardware
Random Forest	n_estimators=200, max_features=log2, max_depth=None	RandomizedSearchCV 20 iter / 3-fold	CPU
XGBoost	n_estimators=500, lr=0.1, max_depth=6, subsample=1.0, $\gamma=0.2$	RandomizedSearchCV 20 iter / 3-fold	CPU
SVR (RBF)*	$C=500$, $\gamma=0.1$, $\epsilon=1.0$	RandomizedSearchCV 10 iter / 3-fold	CPU
MLP	hidden=(128,), ReLU, Adam, lr=0.01, $\alpha=0.01$, batch=1024	RandomizedSearchCV 15 iter / 3-fold	CPU
LSTM	LSTM(64), Dropout(0.2), Dense(32), SEQ=8, lr=1e-3	Manual + Parada anticipada	GPU RTX 3060

3. Analysis of results

This section presents each of the results obtained for each model, along with their evaluation and analysis based on the case study

3.1 Random Forest

On the test set, the Random Forest model gets MAE=1.188, RMSE=3.092, $R^2=0.9979$, and MAPE=4.89%. The MDI feature importance analysis (Fig. 3) shows that previous interval inflows make up 31.1% of the total predictive capacity, which backs up Most of the short-term operational autocorrelation is caused by calendar factors [5], [12]. Next comes passenger exits (14.6%), the inflow/exit ratio (13.6%), and net flow (13.5%). Each calendar variable contributes less than 0.7%. There is no systematic bias because the residual distribution (Figure 3.) is symmetrical around zero (mean = 0.15 pass./interval). Prejudice. The MAE varies by station, from 0.18 at Escuela Militar (low demand) to 4.98 at Portal 80 (high demand) (Fig. 4).

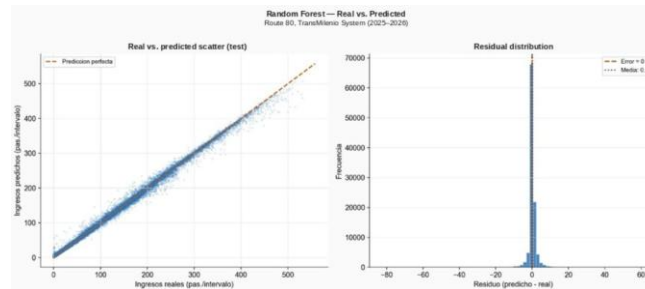


Fig.3. Random Forest (Real vs Predicted) (Own Elaboration)

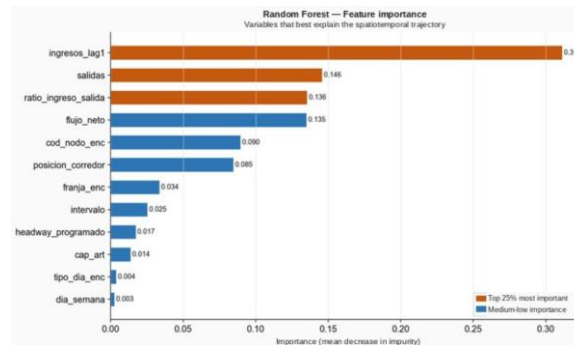


Fig.4. Random Forest (Features importance) (Own Elaboration)

All 13 stations achieved R^2 values greater than 0.995. The model estimated an average residual of 0.15 passengers per interval during the test period from December 2025 to February 2026 (Figure 5), with the previous interval's ridership driving 31.1% of the predictive power, while the calendar

variables each contributed less than 0.7%. Portal 80, as expected, had the highest absolute error, given that it is the busiest node on the corridor.

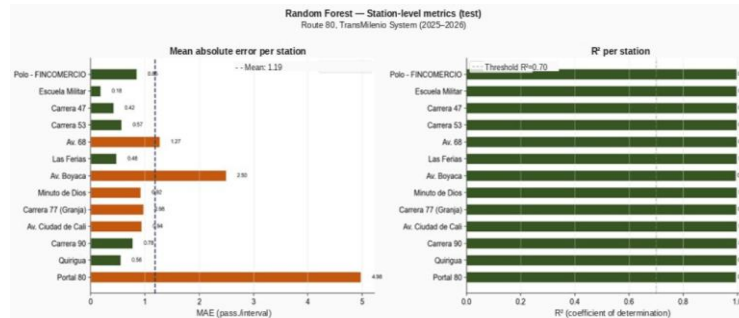


Fig.5. Station Level Metrics (test) (Own Elaboration)

3.2 XGBoost

XGBoost is better than Random Forest on all global measures: MAE=1.013, RMSE=2.526, $R^2=0.9986$, and MAPE=4.07%. We already know that gradient boosting is better than bagging when the demand distributions are different, and this is in line with that[13]. When it comes to feature relevance, Figure 6 shows that prior interval inflows (64.8%) are the most important thing. The normalized gain in RF is 31.1%, which means that the sequential boosting process makes use of short-term autocorrelation. more effectively by assigning consecutive trees to progressively fix residuals that are related to demand modifications. The drop in errors compared to RF is the same for all time periods, with the biggest change. The Mean Absolute Error (MAE) for XGBoost was 1.53 during the PM peak, and the MAE for Random Forest was 1.81. By station, Avenue. Boyacá is the most obvious better (MAE: 1.56 vs. RF: 2.50).

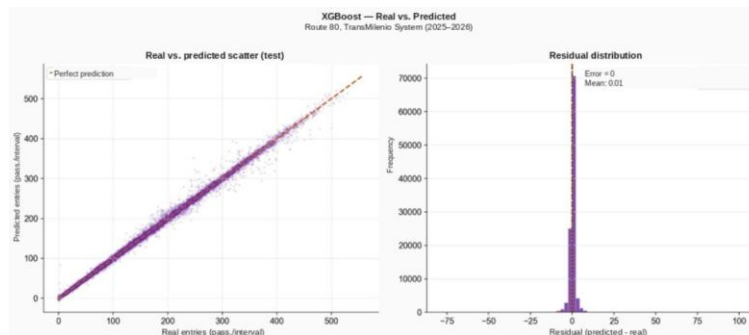


Figure 6. XGBoost: The difference between the actual and expected spread and how the residuals are spread out. There are 0.01 events on average in each interval. (Own Elaboration)

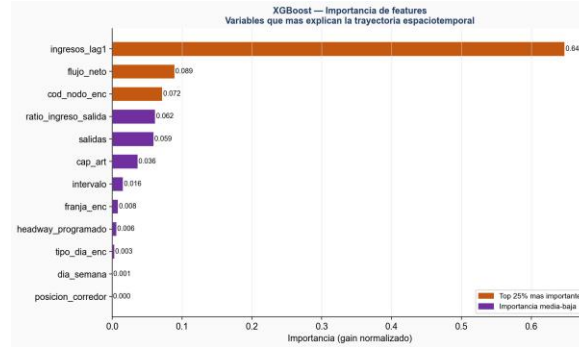


Fig.7. XGBoost: A normalized gain of feature importance. The previous interval inflows were 64.8%, which means that RF used temporal autocorrelation more than RF did. (Own Elaboration)

3.3 Support Vector Regression

Model trained on 80,000 data points with parameters $C=500$, $\gamma=0.1$, and $\varepsilon=1.0$ gets a Mean Absolute Error (MAE) of 0.449, a Root Mean Square Error (RMSE) of 0.584, and a R^2 value of 0.99992. The MAPE, or Mean Absolute Percentage Error, is 3.88%. SVR does better than both RF and XGBoost on all metrics, even though it only uses 23% of the full training dataset standards. This result shows that the RBF kernel is good at showing smooth, non-linear demand relationships in the normal twelve-dimensional feature space [14]. The residuals' distribution (Figure. 9) is almost Gaussian and centered at (-0.04 pas./interval) (Figure 8.), which means there is no clear bias. $R^2 \geq 0.997$ is correct for 12 out of 13 stations. The low value at Escuela Militar ($R^2=0.997$) is because its low-demand profile changes a lot compared to stations (Figure 9.).

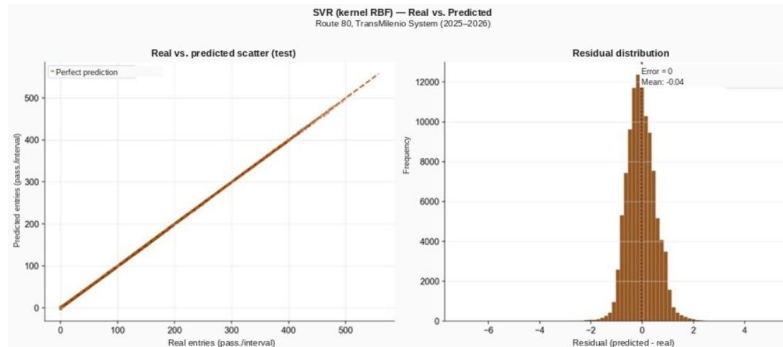


Fig.8. Actual vs. predicted dispersion (Own Elaboration)

The first three models (Figure. 9) show that SVR does better than the others at all times, especially in the morning rush hour, the MAE model has a mean absolute error (MAE) of 0.50, XGBoost has a mean absolute error of 1.29, and Random Forest (RF) has a mean absolute error of 1.54.

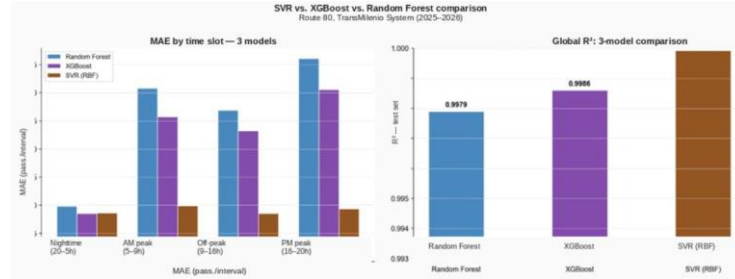


Fig.9. Mean Absolute Error for SVR, XGBoost, and Random Forest (Own Elaboration)

3.4 Multilayer perceptron (MLP)

The MLP design has 128 hidden neurons and uses ReLU activation, Adam optimization, and a learning rate of 0.01. It converges in 23 iterations with early stopping. The best overall performance in this study was MAE=0.011 pass/interval, RMSE=0.043, $R^2=0.9999996$, and 0.089% of the Mean Absolute Percentage Error (MAPE). The MLP is the most efficient model because it only has 1,664 trainable parameters [15] [16]. The model architecture shows that generalization is not compromised, as the training and validation losses are almost perfectly aligned. The predicted versus actual values (Fig. 10) demonstrate good performance with demand ranging from 0 to 573 trips per interval, without any systematic bias. For all static models, the MLP captured between 31% and 65% of the predictive power due to the previous interval inputs, and 11 out of 13 stations achieved an R^2 of 1.000.

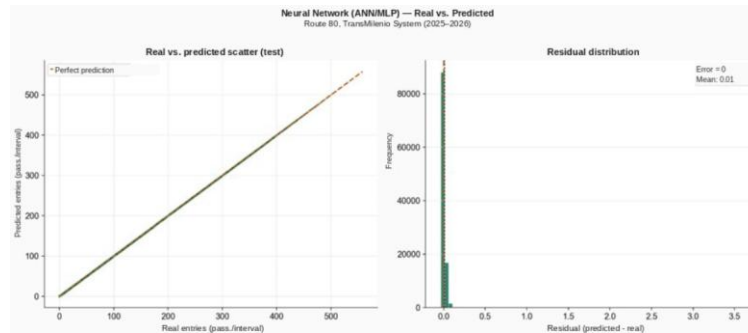


Fig.10. Real vs. predicted, dispersion virtually perfect across the entire demand range (0–573 pass/interval). Average residuals: 0.01 pass/interval (Own Elaboration)

The model's highest Mean Absolute Error (MAE) at 0.026 pass/interval is shown in Portal 80. This is not important for operational purposes.

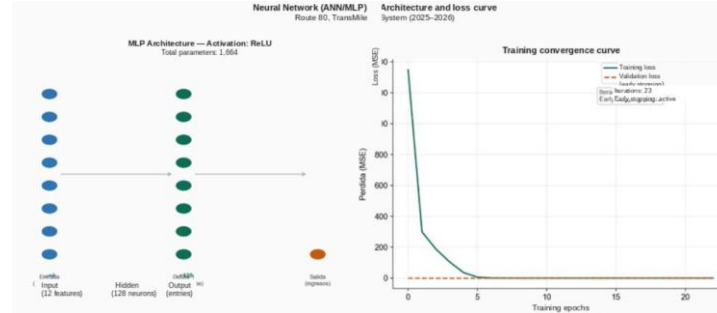


Fig.11. There are 12 inputs, 128 hidden neurons that use ReLU activation, and one output in the MLP architecture. It also has the curve for training convergence. (Own Elaboration)

3.5 Long Short-Term Memory (LSTM)

With an RTX GPU and PyTorch 2.5.1 [17], we made the LSTM model (64 units, Dropout=0.2, Dense(32), SEQ=8). The model speeds up by 3060 and converges in 91 epochs (8.73 minutes of training time). The Mean Absolute Error (MAE) is 11.224 and the Root Mean Square Error (RMSE) is 22.915 $R^2 = 0.884$ and MAPE = 64.64%. The loss drops a lot in the first 20 epochs on the training curve. It stays the same after that the training and validation metrics came together, and the difference was less than 3%. This meant that overfitting was not an issue. The persistent undervaluation of demands surpassing 200 pass/interval (Figure. 12) signifies a structural inconsistency in the issue Formulation: static models use the whole 12-variable vector for the target interval, which includes variables from the past.

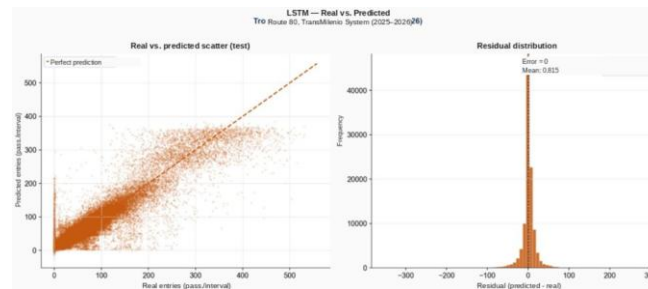


Fig.12. LSTM – Real Vs Predicted (Own Elaboration)

Without seeing what is happening to the target instant, the LSTM is guessing what the inflows will be at time $t+1$ based only on the 8-interval inflows-before t , which makes it very challenging to understand what time it is. The fast inflow changes at Portal 80 during high demand occur so quickly that the 2-hour prediction cannot keep up. This is the historical context that explains the numerous errors at the busy stations, which is why the Mean Absolute Error (MAE) is between 3.61 and 36.09 passengers per interval. Although the LSTM is ranked lowest in global metrics, it does achieve an impressive score of 0.884 on R^2 , demonstrating predictive capability and indicating that it can find time-dependent patterns without using explicit lag variables. This makes

it the default baseline model for any structural enhancements based on attention or bidirectional mechanisms [13], [18].

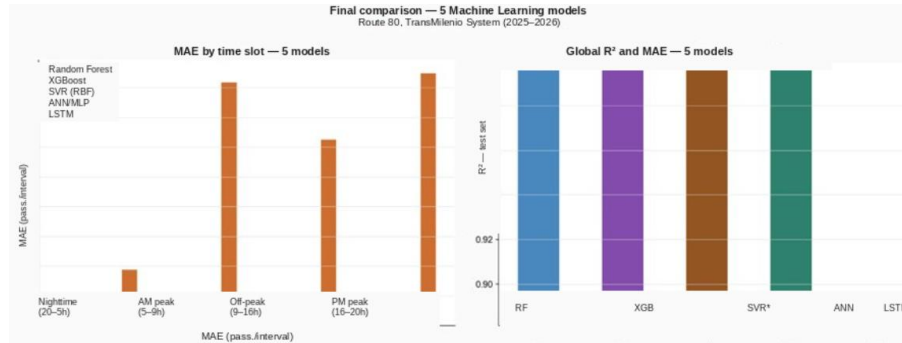


Fig.13. The last five models are compared by their Mean Absolute Error over time and their global R^2 /Mean Absolute Error. Final panel of the comparison study. (Own Elaboration)

3.6 Computational cost

Tests were conducted on an Intel Core i9-10900X (3.70 GHz, without a GPU). The following metrics were recorded: training time, dataset size, full inference on 107,163 test records, latency (ms) per observation, and real-time tuning (900 seconds, 15-minute window). All static models met the real-time targets. SVR (the slowest, on 80,000 records) trained in 17,363 ms, within 2% of the window. LSTM required a GPU (RTX 3060) to be viable; XGBoost inference was not comparable due to the network, but was expected to perform well.

For a 13-station route with 15-minute intervals, the static models handled the workloads well. Scalability to the full TransMilenio network (over 140 stations, over 15 routes) implies approximately a 5-fold increase in network load, along with the need for regular retraining, increased pipeline robustness, and fault tolerance. MLP is the best option for deployment across the entire network: it has the fastest training time (43 seconds on CPU), the lowest latency per observation (4.8 μ s), and does not require a GPU.

Table 3. Metrics for the Test Set (December 2025 –febrero 2026) to use as a reference. *SVR learned from 80,000 records. (Own Elaboration)

Model	MAE	RMSE	R^2	MAPE (%)	Parameters	Hardware
Random Forest	1,188	3,092	0,9979	4,89	200 trees	CPU
XGBoost	1,013	2,526	0,9986	4,07	500 stimators	CPU
SVR (RBF)*	0,449	0,584	0,99992	3,88	4.360 SVs	CPU
MLP	0,011	0,043	0,9999996	0,089	1.664	CPU
LSTM	11,224	22,915	0,8844	64,64	22.081	GPU RTX 3060

4. Discussion

The results corroborate the four analytical conclusions and prove that the methodological decisions taken and how they would influence the execution of the project are evident.

MLP is preferable as it considers inflows from the prior interval as part of the input vector for the subsequent interval. The autoregressive variable accounts for 31–65% of the predictive capability of all static models, and MLP leverages this data due to a universal approximator that can model any non-linear relationships [3],[16]. This result matches the findings from studies on short-term transport demand forecasting, emphasizing that recently operational state parameters hold greater importance than the calendar [5]. It also exerts a direct effect on how it functions: for any operator intending to apply short-horizon demand forecasting on a BRT route, the most crucial data engineering decision is ensuring that the inflow count for the most recent period is available.

Fourth, XGBoost is always better than RF for all stations and all time periods. This validates the assertion that sequential boosting outperforms bootstrap aggregation when the target variable is not uniformly distributed. This is a natural segment of a corridor where Portal 80 receives inflows every interval that are up to 10 times higher than those of Escuela Militar [4]. XGBoost's regularization inhibits trees from overly fitting to the demand trends of a specific locality. The RF method of averaging to reduce variation fails to perform in scenarios where demand is highly variable across different stations.

Third place goes to SVR, despite it training only 23% of the data research confirms that the RBF kernel models the smooth non-linear demand relationships in the standardized feature space [14]. The benefit of sampling in this case is attributed to the nature of SVR that provides a predictive function determined solely by support vectors irrespective to the size of the training data. The problem of using a training data of n size in the systems with a lot of data is that the training takes $O(n^2)$ time [19].

In a temporal and spatial perspective, the fourth point, looking at the errors, the discrepancies among the spatial stations are greater compared to the discrepancies across the temporal intervals for all models. It indicates that the topological characteristics of the corridor, in terms of relative position, as a collecting or distributing node, and demand density, are more influential on the predictive challenge than the time of the day cycle [20], [21]. The explanation for high MAPE for all models during nighttime is related to the presence of some denominators which are close to zero and this does not indicate that the predicted values are not relevant [22]

5. Conclusions

This study offers the first fully documented analysis of five machine learning models from three different algorithmic families, applied to the spatiotemporal prediction of passenger demand, that TransMilenio BRT Trunk Line 80, using a fine-grained operational dataset, and under a rigorous time-frame validation to mimic real-world implementation circumstances.

Of the static models, the MLP proved to be the most successful in real time settings (MAE = 0.011, $R^2 = 0.9999996$, 4.8 μ s/observation), and only required CPU processing and 1,664

parameters, which allowed it to be operationally deployed in real time settings, without specialized processing units. XGBoost provided a highly accurate prediction ($R^2 = 0.9986$) and offered insightful and interpretable classification of features through normalized gain, which makes it the most suitable choice for operational diagnostic and explainable AI tasks. SVR operated successfully with smaller amounts of historical data ($R^2 = 0.99992$) and offered a correct and viable choice when data records were scarce—it was applied with proper input scaling.

In view of the above, LSTM performed poorly when compared with static models, but this is not a result of the model architecture. However, it is a result of a more complex prediction and a higher-order task in his study, which LSTM is trying to predict passenger demand without the data of the operational state of the target interval. This is a methodological distinction that should be noted in this study for the sake of clarity and should not be viewed as a negative aspect of recurrent architectures. Despite this, his R^2 of 0.884 is indicative of a significant level of performance given the challenges of unprocessed sequential input data.

The first operational conclusion centers on the prediction rank of the lagged input stream from previous intervals. In all of the static models, this feature was ranked between 31% and 65% relative importance and, thus, shifted the primary engineering challenge from model selection to the quality of the data stream. The second conclusion notes that the model predictions during the morning and afternoon peak periods exhibit a systematically higher prediction error. This is a consequence of demand volatility that is beyond model design and requires the incorporation of additional data inputs like weather data, incident telemetry, and fleet-level data streams. The third conclusion notes that, in all cases, the intra-station variance exceeded the inter-window variance. In terms of demand, Portal 80 is the terminal that yielded the greatest absolute error in the network along with being the terminal with the highest demand.

Taking this into consideration, from an operational perspective, the MLP–XGBoost combination provides the optimal operational performance, with MLP designed for real-time low-latency inference capabilities and XGBoost designed to provide feature-level explanations and support anomaly detection. This combination provides the ability for transit operators to manage trade-offs between the predictive performance and level of explainability demanded by regulatory and planning stakeholders.

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